

ECSE 4660/6660: Internetworking of Things

Lecture 7: PHY Layer Fundamentals – Part V

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ECSE, RPI

Spring 2026

Administrative

- Use Piazza for Lab questions
 - <https://piazza.com/rpi/spring2026/ecs/e46606660>
- Today's Quiz is available on **LMS**
 - Multiple choice questions
 - Due tonight 11:59 PM
- Project group formation **due 1/30!!**
 - Add Project Group here [IoT_SP26_project.xlsx](#)
- Project Proposal submission **due 2/6**
 - Template Provided
 - Rubrics Provided
 - Request extra office hour or visit my office JEC 6018
- HW1 on **LMS** **due 2/6**
 - No Late submission allowed
- Lab is located in JEC 5312
 - Access is provided to all registered students
 - You can visit lab anytime Mon-Fri 7AM-10PM
 - Labs are done **individually**- Plagiarism penalized heavily (See Syllabus pdf)
 - Late-submission Policy
 - **TA hours 4-5 PM** daily in 5312 lab
 - Lab 2 due **Fri, 2/13**

Why Measure and Model Networks

Ellen Zegura

Regents and Fleming Chair Professor, School of
Computer Science
Georgia Tech

Mercer Distinguished Lecture Series

LOW 3051

Wed, February 11, 2026 at 4:00 PM

In this talk, I will highlight three examples of network measurement and modeling projects that span my 30-year career. The first project advanced Internet topology modeling from simple random-graph based structures to incorporate hierarchy and locality, and spurred measurement efforts to better understand Internet structure. The second examined performance of the Domain Name System (DNS), a key service in the operation of the Internet since the mid-1980s. The third represents a five-year effort from research prototype to FCC-approved cellular measurement application and infrastructure, with the potential to impact mobile broadband investments. Throughout I will share lessons learned about research, development, funding, and working with students.

Dr. Ellen Zegura is Regents and Fleming Chair Professor in the School of Computer Science at Georgia Tech. In computer networking, she is known for her work on the GT-ITM suite of Internet topology tools, which were used for 20 years after initial release. In mobile wireless networking, she and colleagues invented the concept of message ferries to facilitate communications in environments where network connectivity is unreliable and/or sparse. Her work in computing and social good includes projects in Liberia with the Carter Center, with Native Americans in Southern California, and with residents of the Westside communities of Atlanta. She is currently on loan from Georgia Tech to the National Science Foundation where she serves as Acting Assistant Director for Computer and Information Science and Engineering (CISE). She served on the board of the Computing Research Association, including as Board Chair from 2019-2021. She is a Fellow of the IEEE and a Fellow of the ACM.



Wireless Transmission: Outline

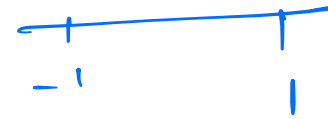
1. Modulation Schemes (Bits to Symbol) 
2. Error Detection and Correction
3. Multi-path Fading
 - Introduces ISI
4. OFDM (WiFi and Cellular)
 - Mitigates ISI, high rates, long range

Digital Modulation

OOK

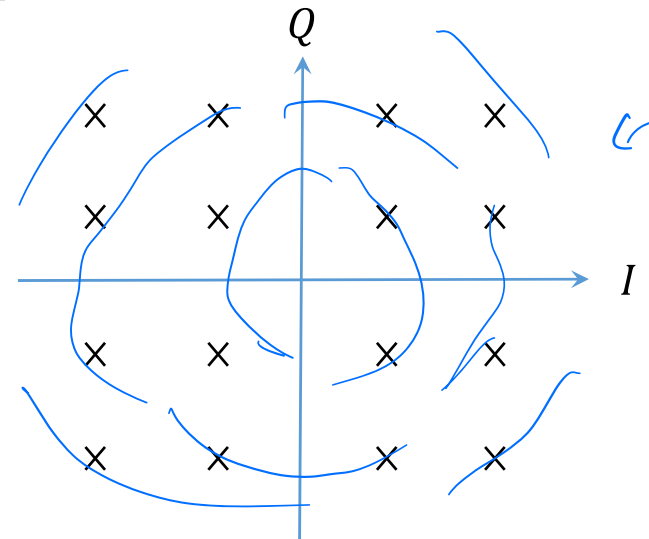
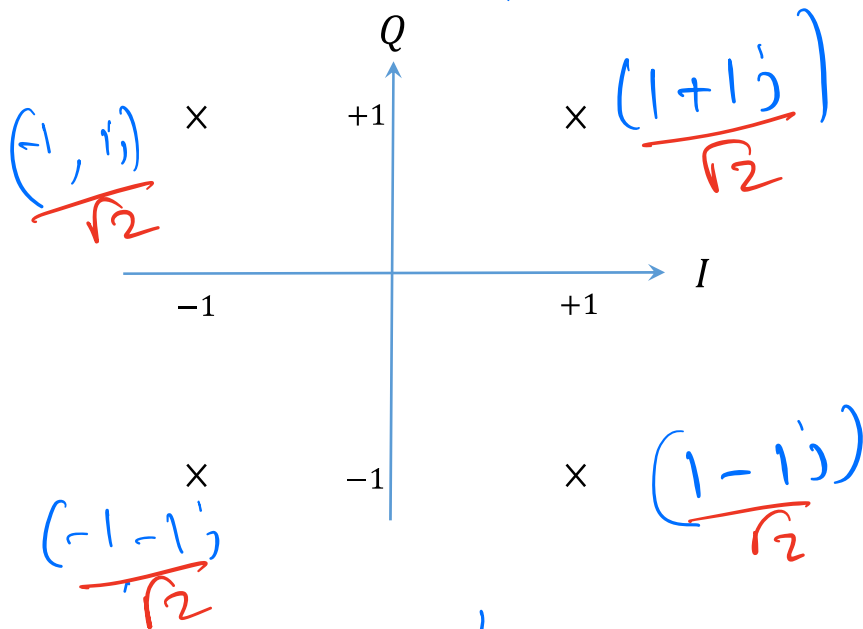


BPSK



Power/sym = 1

- QAM : Quadrature Amplitude Modulation
 - Scale to maintain total radiated power



4 QAM / QPSK

$$\begin{aligned} \text{total power} &= \left(\frac{1}{\sqrt{2}} \right)^2 + \left(\frac{1}{\sqrt{2}} \right)^2 + \dots \\ &= \left(\frac{1}{\sqrt{2}} \right)^2 + \left(\frac{1}{\sqrt{2}} \right)^2 \times 4 \leftarrow \text{symbols} \\ &= 4 \end{aligned}$$

16 QAM

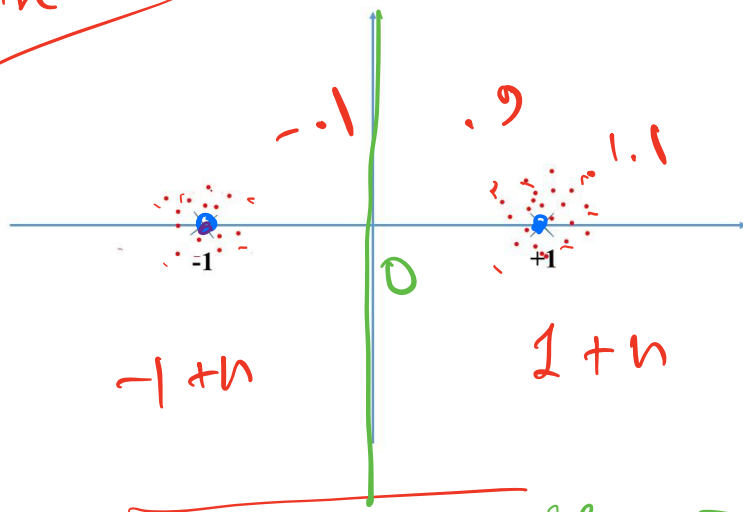
power per symbol
P/sym = 1

Additive White Gaussian Noise

AWGN
additive
white
Gaussian noise

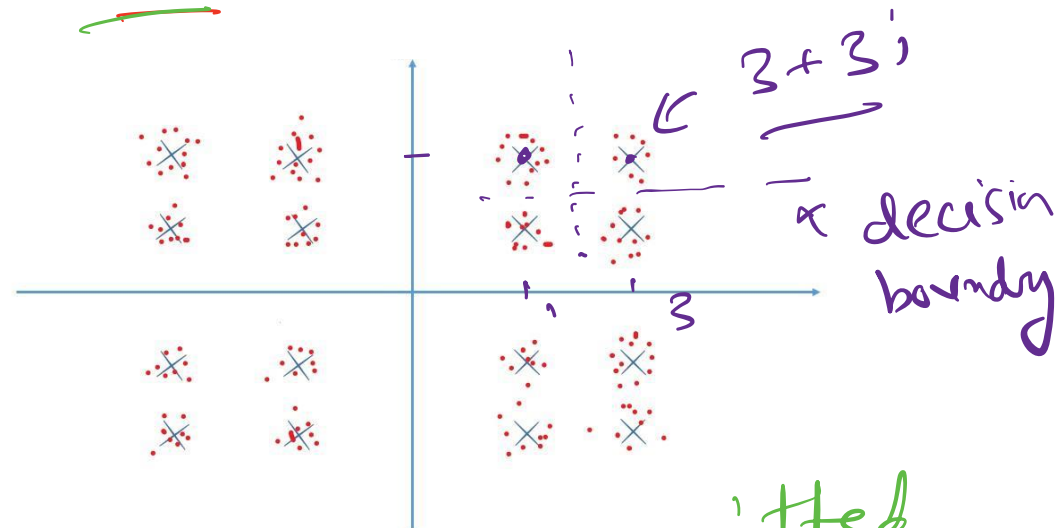
BPSK: 1 bit per symbol

$$y = x + n$$



if $y > 0 \Rightarrow$ bit 1 is transmitted
if $y < 0 \Rightarrow$ bit 0 is transmitted

16 QAM: 4 bits per symbol



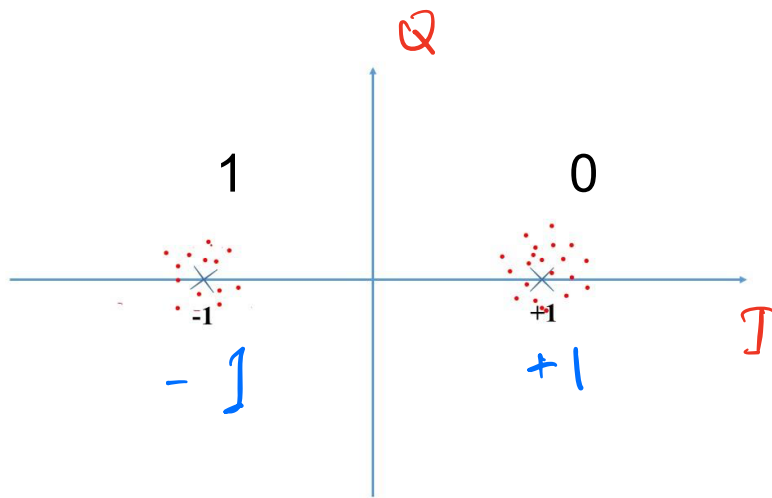
Maximum Likelihood Detector $\rightarrow$ Decision Boundary

min. $L_{\text{error}} = \begin{cases} P(\text{bit 1 received} \mid \text{bit 0 is transmitted}) \\ + P(\text{bit 0 received} \mid \text{bit 1 is transmitted}) \end{cases}$

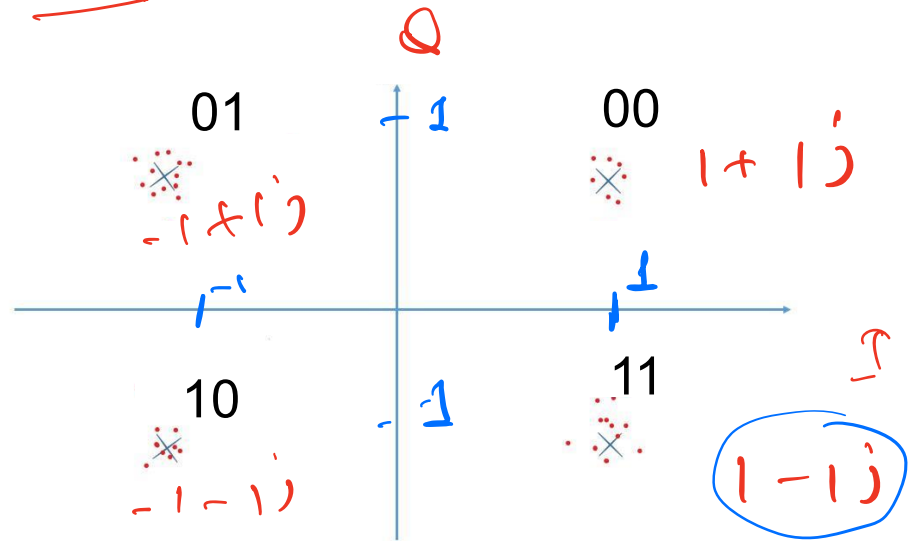
How to map the bits to constellation points?

$00 \text{ } \textcircled{11} \text{ } 00$ bits $\rightarrow +1, +1, -1, -1, +1, +1$ symbols $\rightarrow 1+j, 1-j, 1+j$ 3 symbols

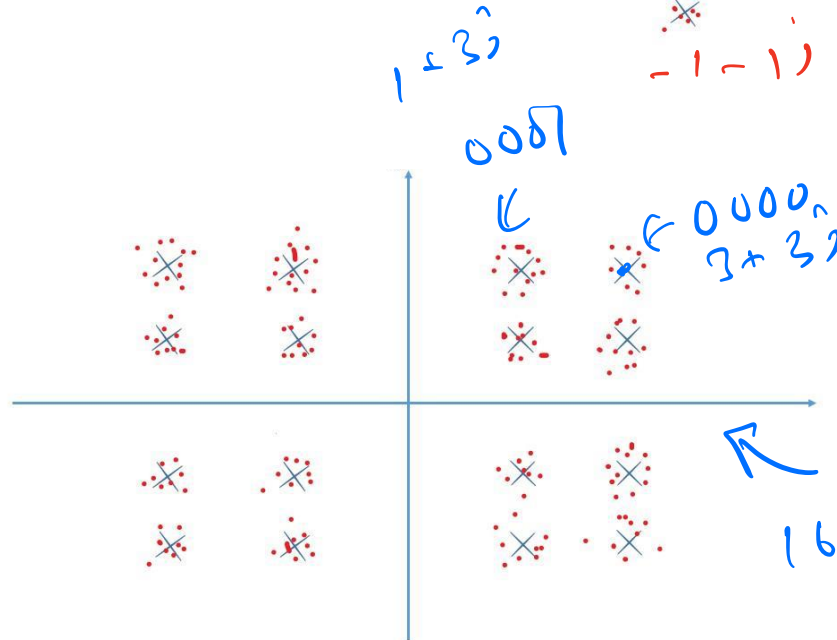
BPSK: 1 bit per symbol



QPSK/4 QAM: 2 bits per symbol



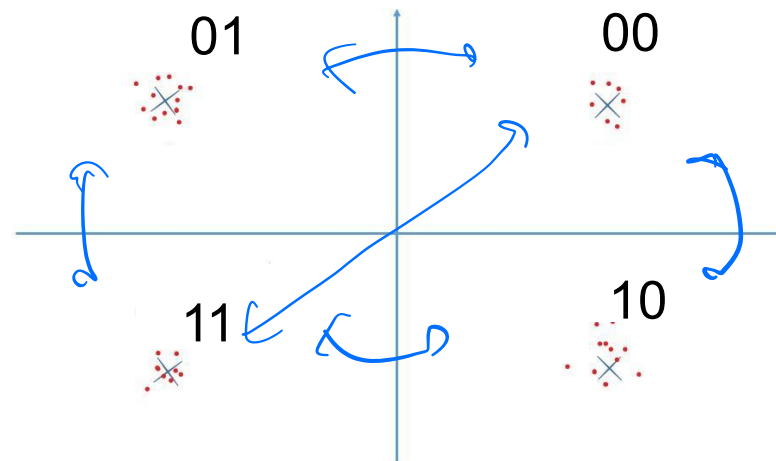
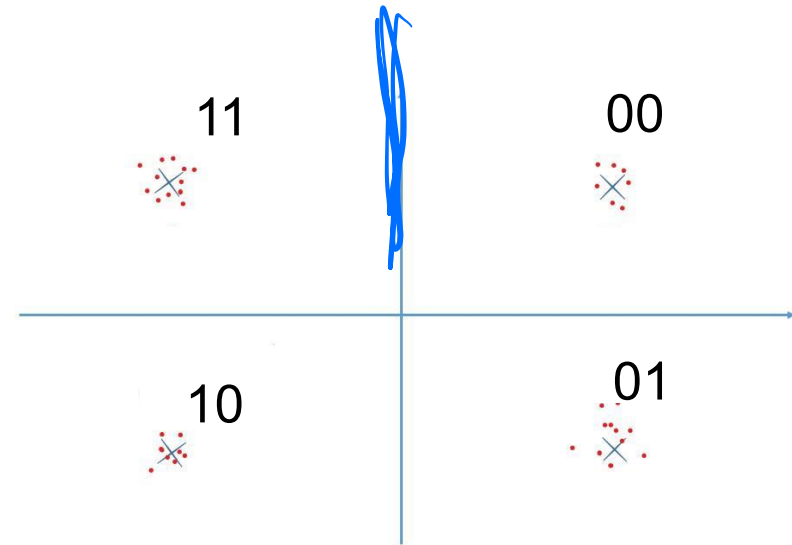
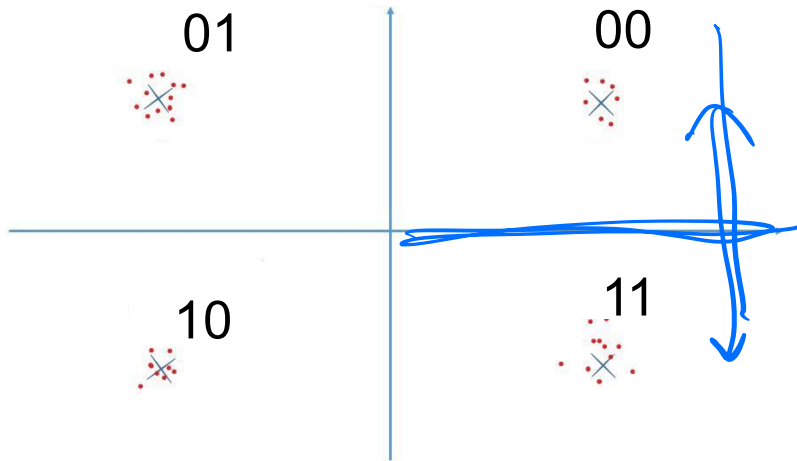
$\uparrow$ 2 bits / symbol



$\uparrow$ 16QAM $\Rightarrow$ 4 bits / symbol

How to map the bits to constellation points?

QPSK: 2 bits per symbol

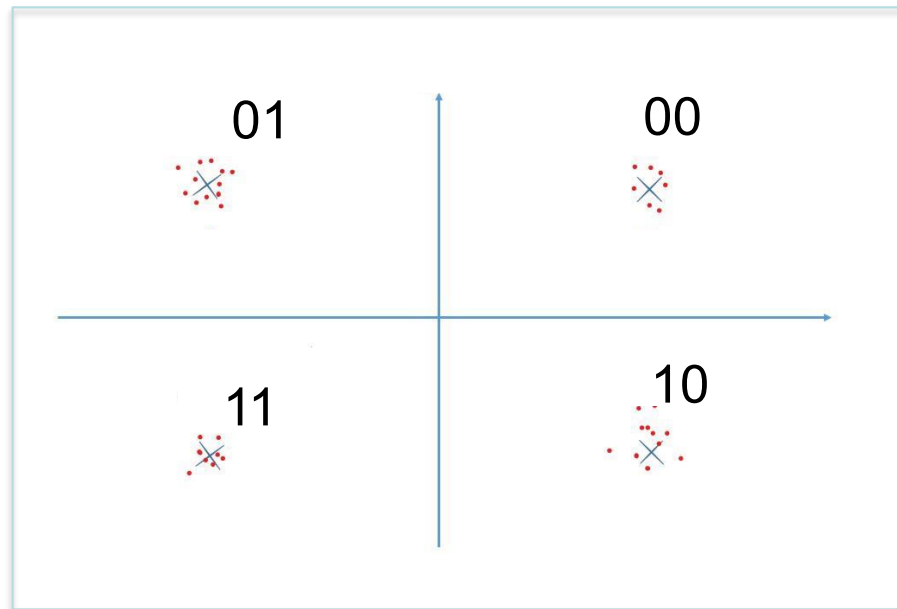
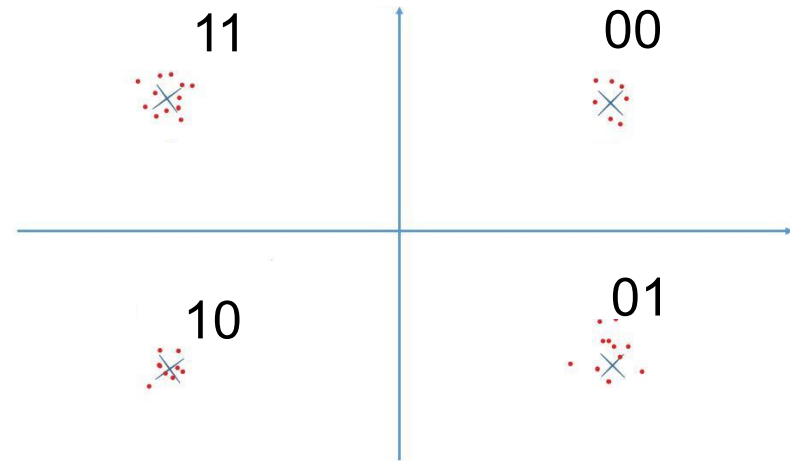
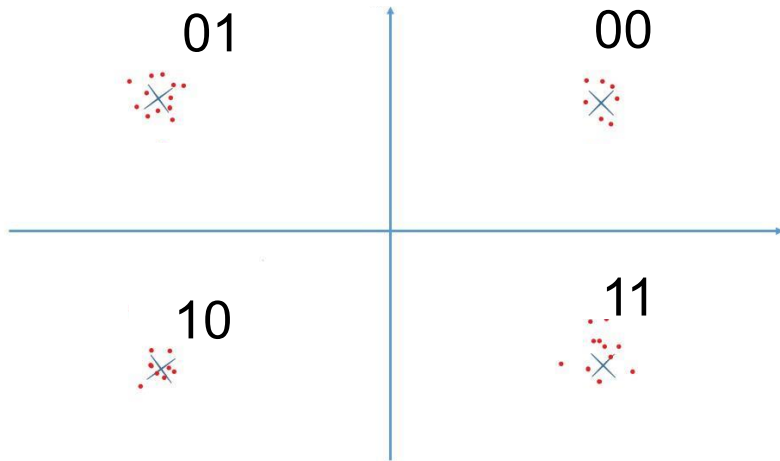


Q: Which of the three mapping is better?

*1 symbol error
⇒ 1 bit error*

How to map the bits to constellation points?

QPSK: 2 bits per symbol



Lab 2

Gray Codes:

Minimize bit error

Adjacent symbols
only differ by 1 bit

General radio.

Bit-Error-Rate

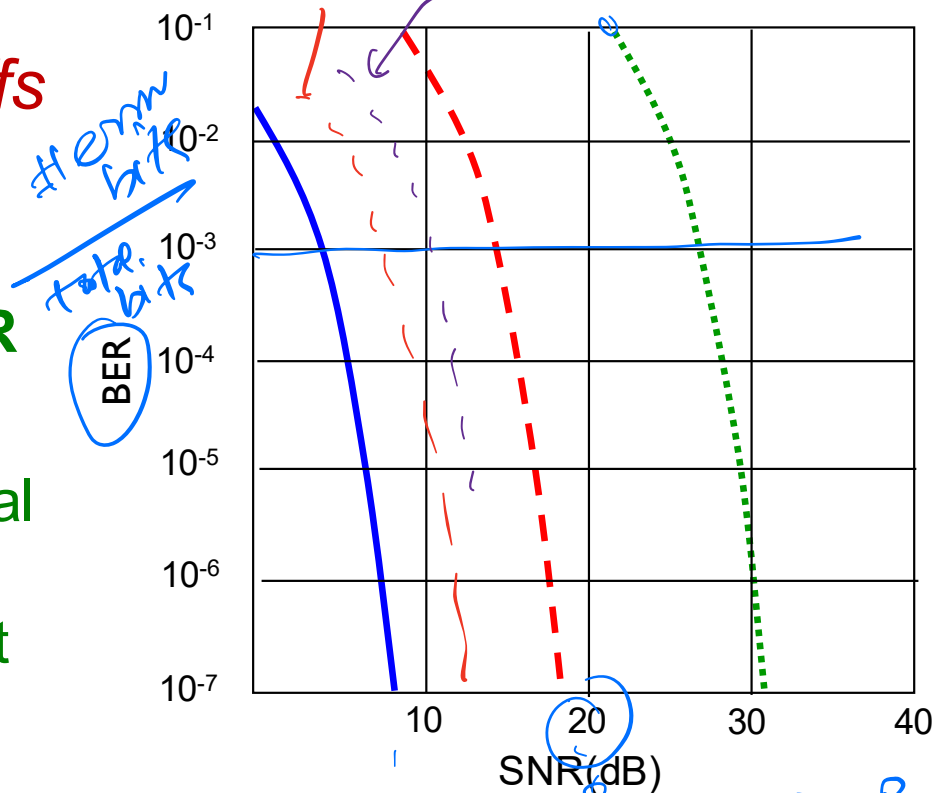
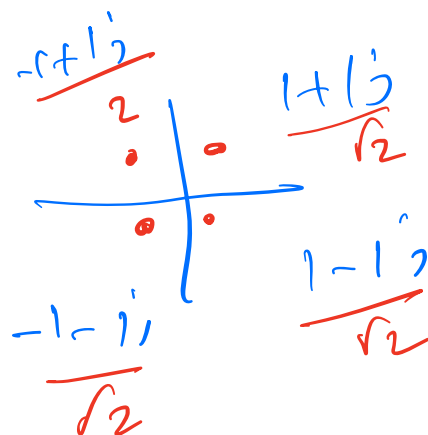
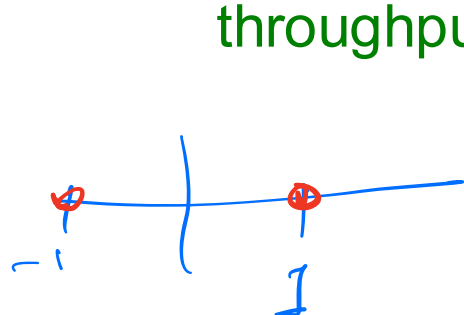
Lab 2

■ SNR versus BER tradeoffs

- given physical layer modulation:

Higher SNR $\rightarrow$ Low BER

- given SNR: choose physical layer that meets BER requirement, giving highest throughput

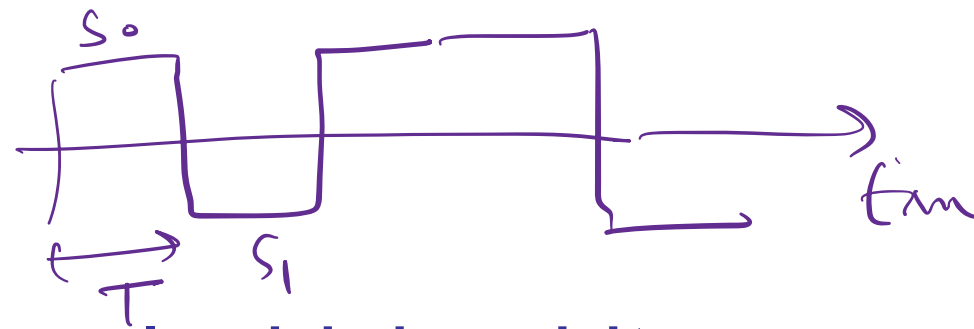


- QAM256 (8 Mbps)
- QAM16 (4 Mbps)
- BPSK (1 Mbps)

QPSK

Power Rx
wire

Higher Bit Rates



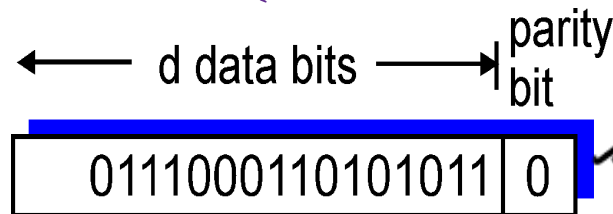
- From the previous example, higher bit rates could be achieved by:
 - Increasing the symbol rate (i.e. decrease T)
 - ✓ Nyquist criterion (to avoid aliasing): pulse rate must be less than $2W$.
 - Increase the number of possible symbols from 2 to $M > 2$ i.e. bit rate is $(\log M)/T$
 - How much can we “pack” while maintaining the same error probability per bit?
 - ✓ Need to increase the power levels (so that there is same degree of separation between the different levels)

Error Detection and Correction

- Add Redundant bit to
- Checksums → Detect Errors

only works if
only 1 bit is
in error

- Parity Check

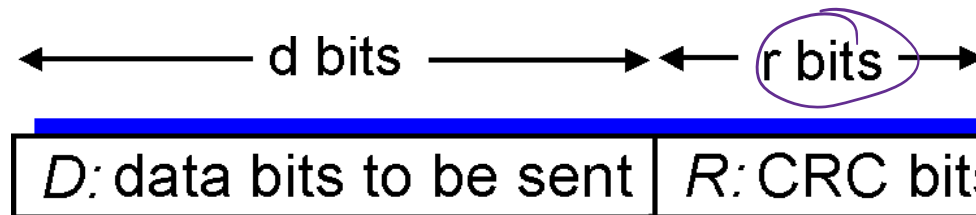


9 ones ⇒ parity is 0

even # of 1 ⇒ ~~1~~
odd # of 1 ⇒ ~~0~~

1 bit error
odd number
of bit flips

- CRC: Cyclic Redundancy Check



bit
pattern

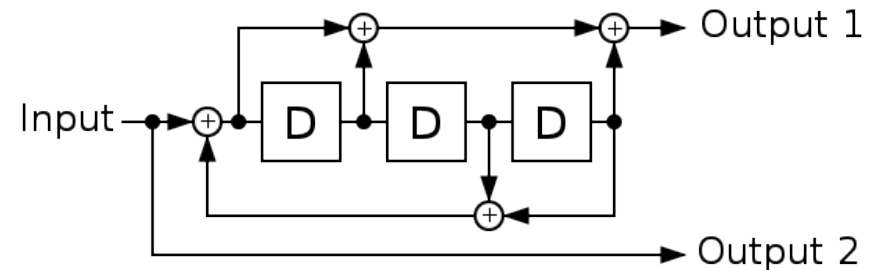
larger r
⇒ detect
more errors

Error Detection and Correction

■ FEC: Forward Error Correction

- ✓ ■ Repetition Code
- ✓ ■ Convolutional codes
- ✓ ■ Reed Solomon codes

- {
 - Turbo codes 4G
 - LDPC codes 5G



- ## ■ Coding Rate

$= \frac{2}{3}$
→
 every two bits add 1 more bit $= \frac{d}{d+r}$

$\uparrow$ coding rate $\Rightarrow$ less correct
 $\downarrow$ coding rate $\Rightarrow$ redundancy $\Rightarrow$ $\uparrow$ correct bits

Lab 2

Gnuradio

Open

File Edit View Run Tools Help

untitled - BPSK_SER_simulation

Run the flowgraph

Options
ID: BPSK_SER_simulation
Generate Options: QT GUI

QT GUI Range
ID: SNR
Label: SNR (dB)
Default Value: 10
Start: -15
Stop: 20
Step: 1

Variable
ID: modulation
Value: <gnuradio...a99cc4e0>

Variable
ID: BPSK_MAP
Value: -1, 1

Import
Import: math

Variable
ID: samp_rate
Value: 320k

Random Source
Minimum: 0
Maximum: 2
Num Samples: 1k
Repeat: Yes

Virtual Sink
Stream ID: GT

Chunks to Symbols
Symbol Table: -1, 1
Dimensions: 1

Throttle
Sample Rate: 320k

Virtual Sink
Stream ID: Transmitter

Virtual Source
Stream ID: Transmitter

Noise Source
Noise Type: Gaussian
Amplitude: 100m
Seed: 42

Add

QT GUI Constellation Sink
Number of Points: 1.024k
Autoscale: No

Virtual Sink
Stream ID: Receiver

BPSK Decoder

Constellation Decoder
Constellation Object: ...0>

Virtual Source
Stream ID: Receiver

Virtual Source
Stream ID: GT

BER
Window Size: 1M

QT GUI Number Sink
Name: Bit Error Rate
Autoscale: No
Average: 0
Graph Type: Horizontal

tx bits

0,1,0,0,1

-1,1,-1,1

tx symb

tx sym

tx bits

noise

display GUI

save

Two blocks you need to write codes for

decoded bits

BER

0PSK with Gray

BER

ENR

Block paths:
/usr/local/share/gnuradio/grc/blocks
/usr/local/share/gnuradio/grc/blocks

Loading: "/home/uone/Gnuradio/gnuradioworkspace/RPI-IoT/BPSK_QPSK/Problems/BPSK_SER_simulation.grc"
>>> Done

running messages, including python code printout

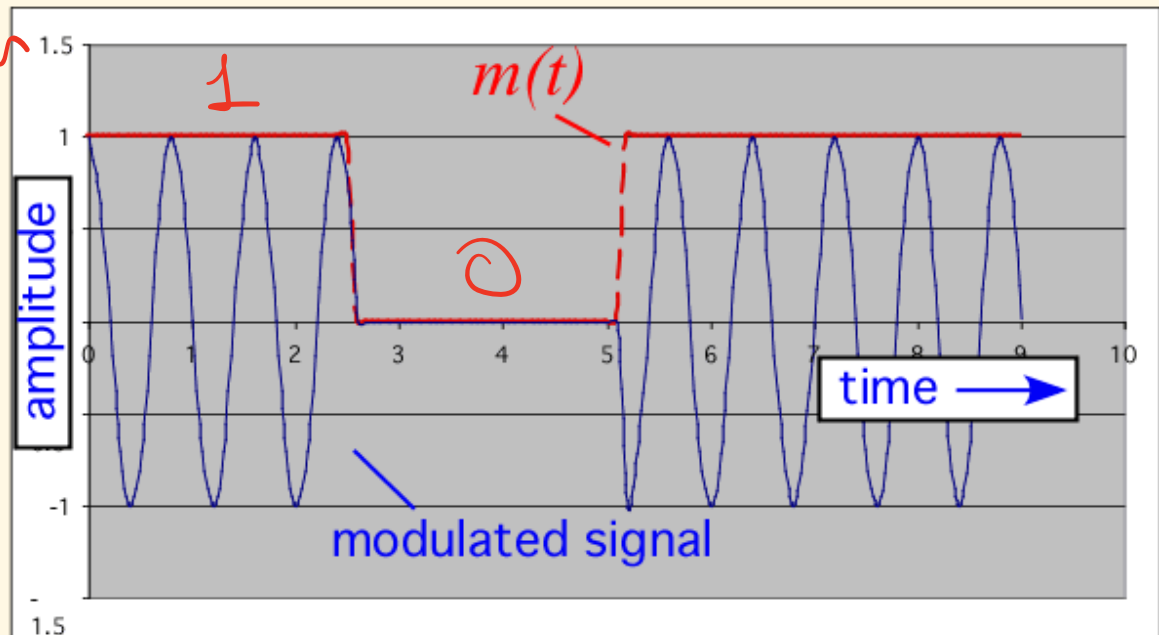
Id	Value
Imports	
import_0	import math
Variables	
BPSK_MAP	-1, 1
modulatic digital.constellation_bpsk()	base()
samp_rate	32e4
SNR	<Open Properties>

On Off Keying ($m(t)=0$ or 1)

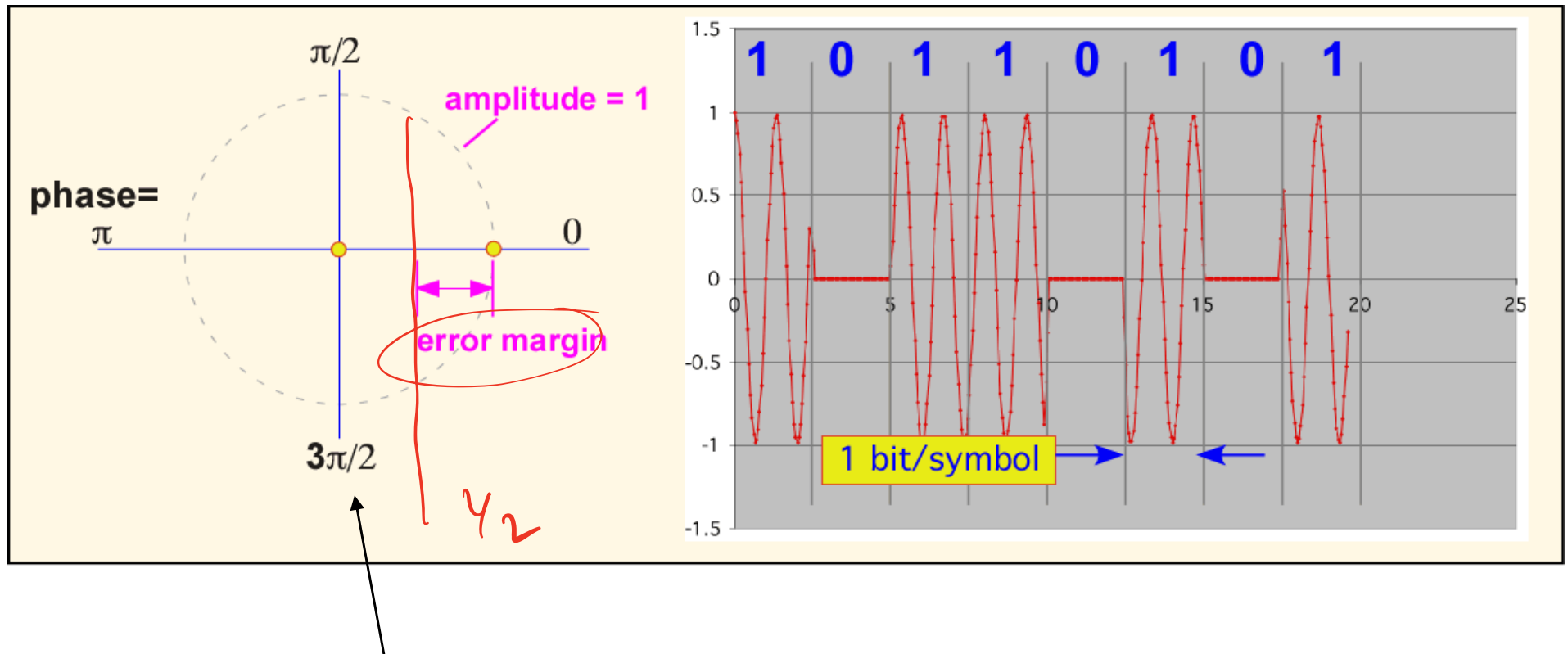
On-Off Keying

$$f(t) = m(t) \cos(\omega_c t)$$

$$m(t) = \begin{cases} +1 \\ 0 \end{cases}$$

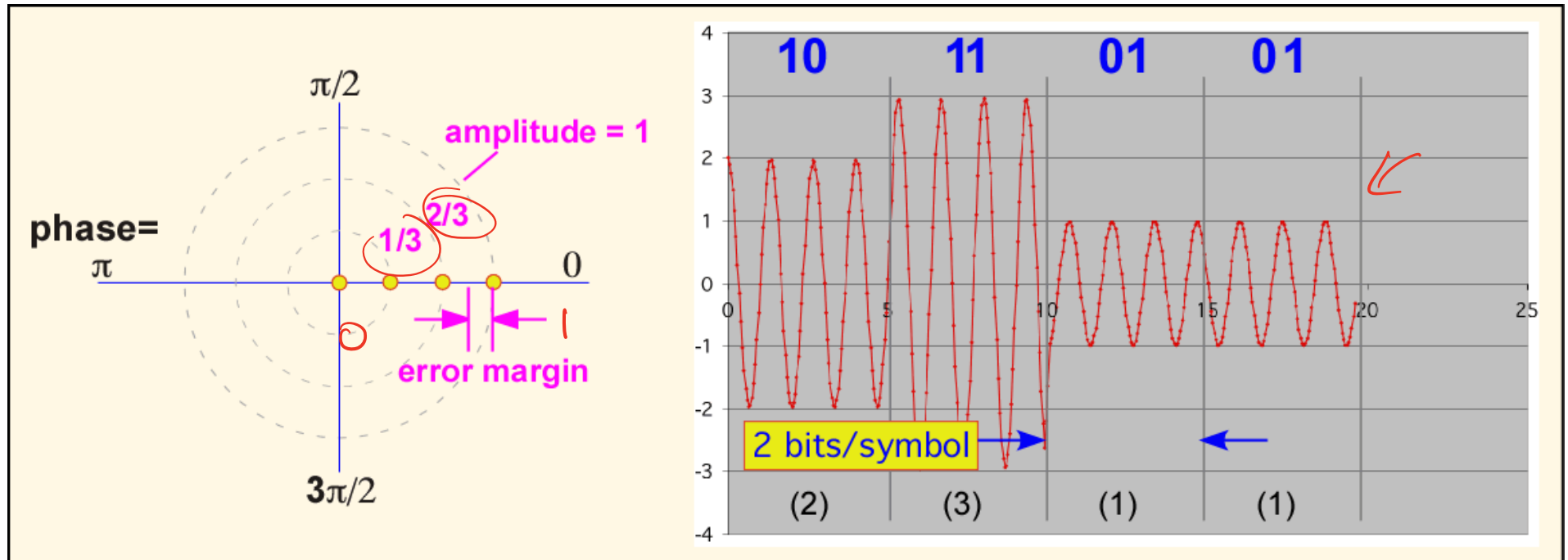


On Off Keying ($m(t)=0$ or 1)



Signal Constellation diagram to compute the probability of incorrect detection (also called **Signal-Space** representation)

4-Amplitude Shift Keying

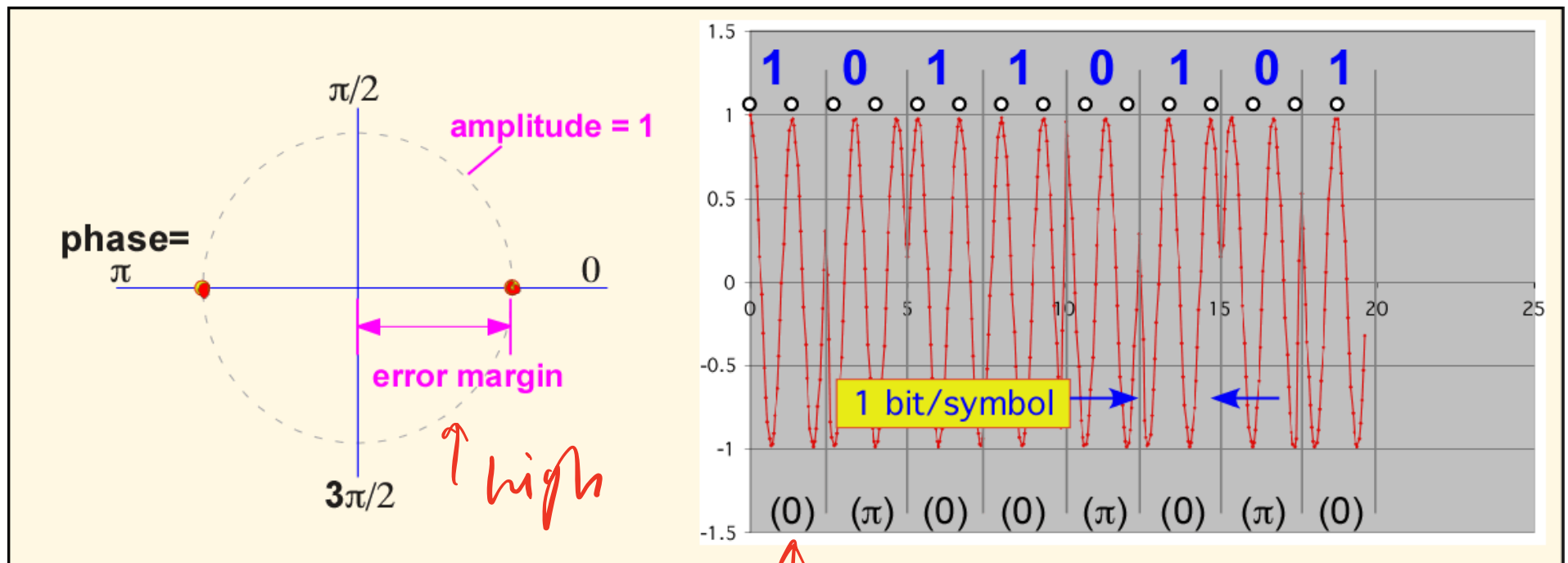


-4ASK transmits 2 bits per symbol

-Provides same data rate with half the bandwidth -(or, provides twice the data rate of OOK with same bandwidth)

-But, error margin is much smaller than OOK

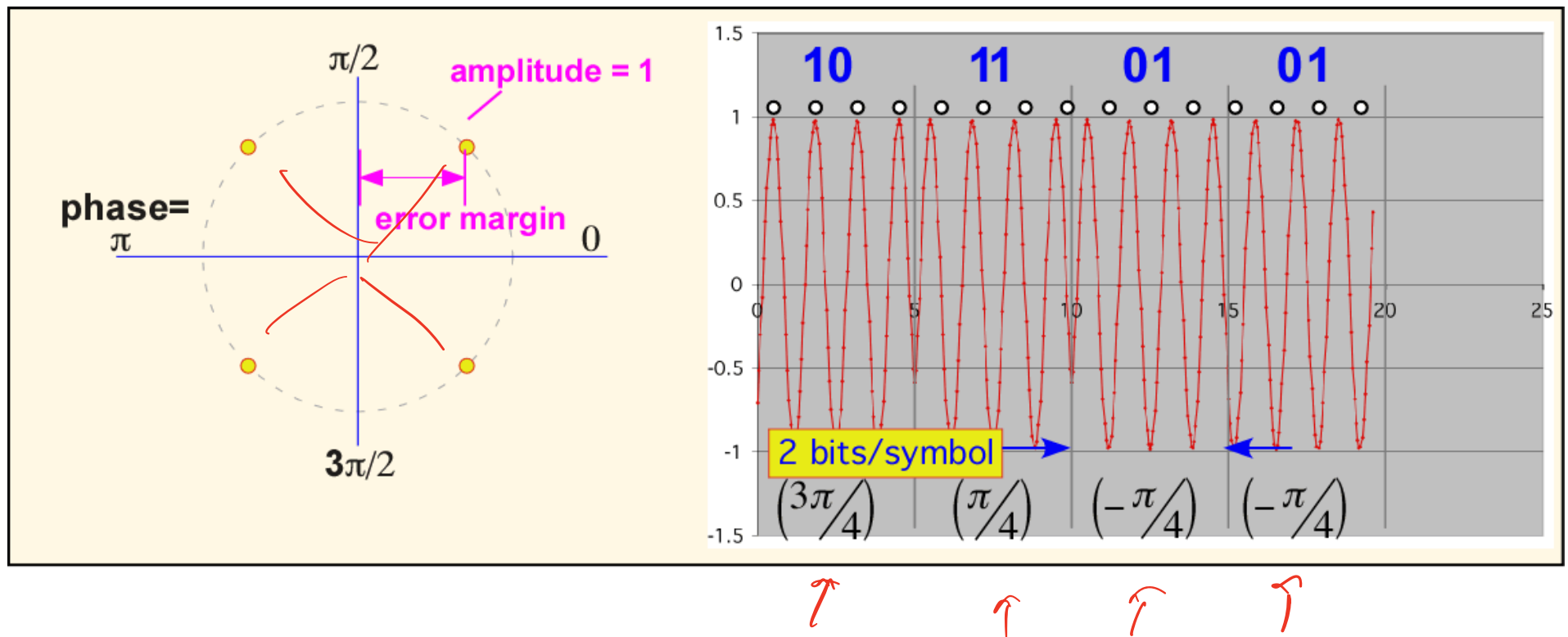
Binary Phase-Shift Keying (BPSK)



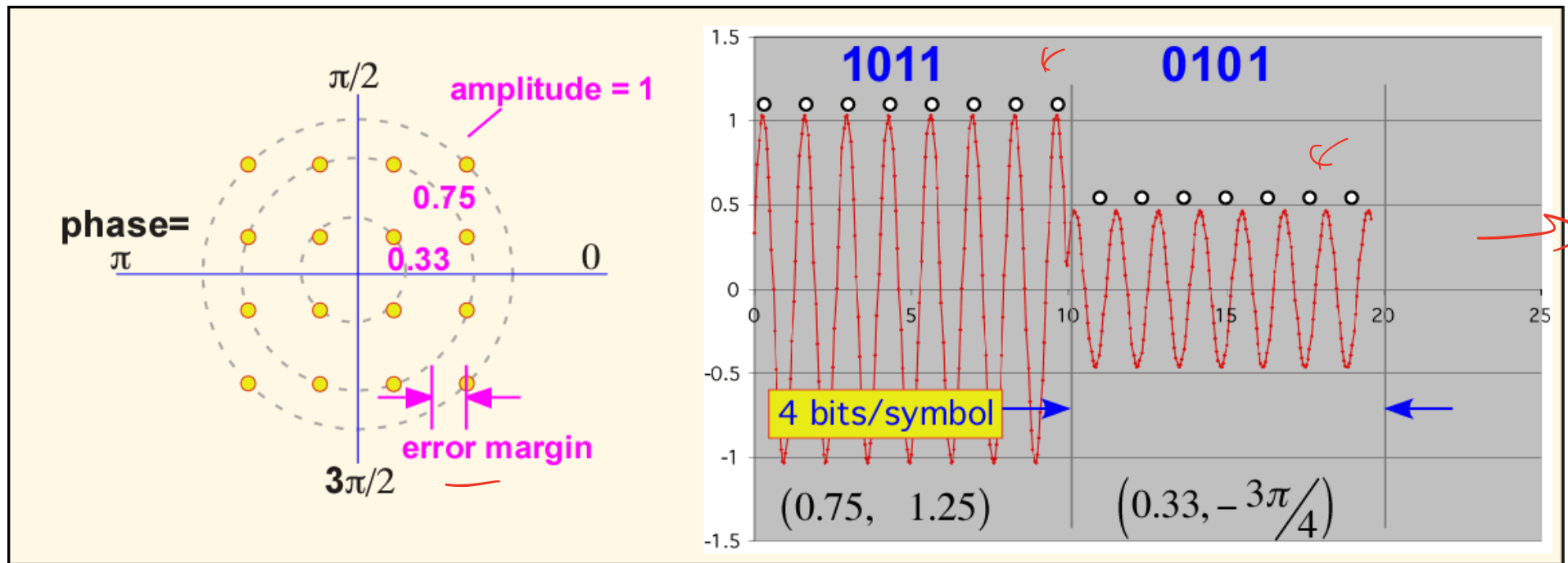
Handwritten notes below the waveform:

- phase $\uparrow$ phase π
- $(\cos 2\pi f_c t)$
- $(+1)$
- $\cos(2\pi f_c t + \pi)$
- $= -\cos(2\pi f_c t)$
- (-1)

Quaternary Phase Shift Keying (QPSK)



16-Quadrature Phase Shift Keying (16QAM)

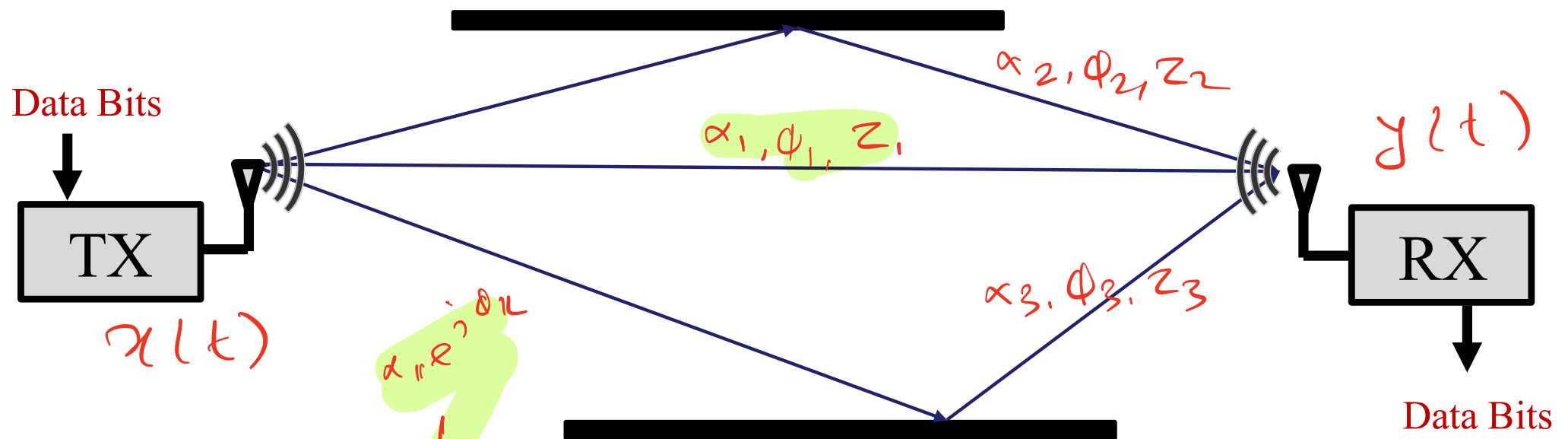


-A combination of amplitude and phase modulation, trying to keep the points in the constellation equally spaced from each other.

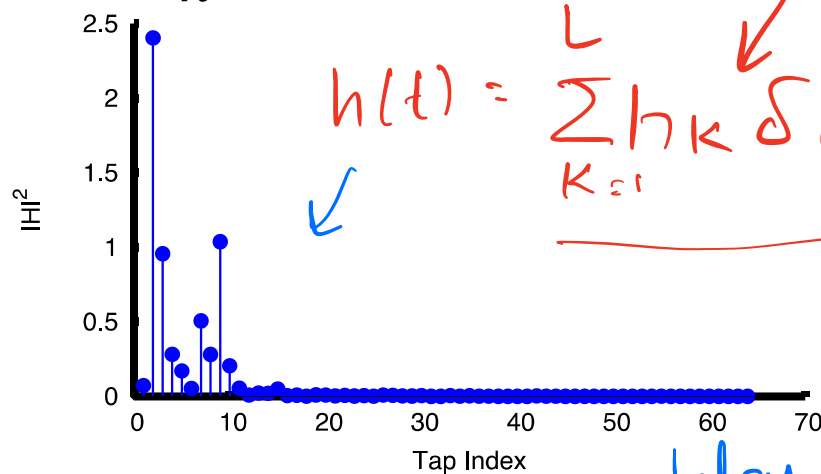
Wireless Transmission: Outline

1. Modulation Schemes (Bits to Symbol)
 2. Error Detection and Correction
 3. Multi-path Fading
 - Introduces ISI
 4. OFDM (WiFi and Cellular)
 - Mitigates ISI, high rates, long range
- 

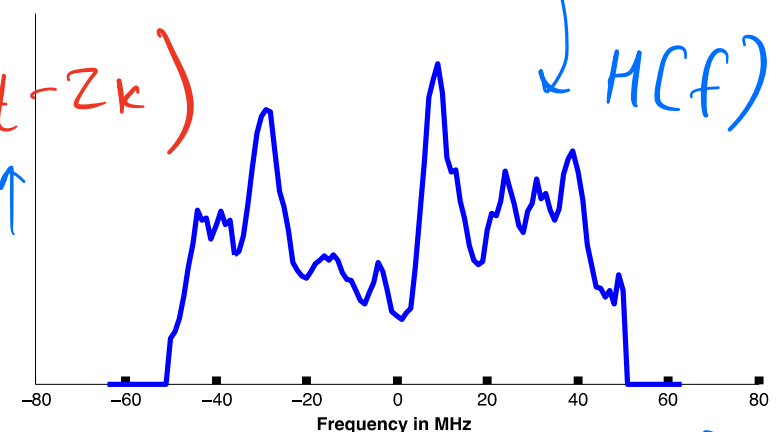
Wide Band vs Narrow Band Channel



$$\underline{y(t)} = \sum_k h_k x(t - \tau_k) = \underline{h(t) * x(t)} \xLeftrightarrow{\text{FFT}} \underline{H(f)X(f)}$$

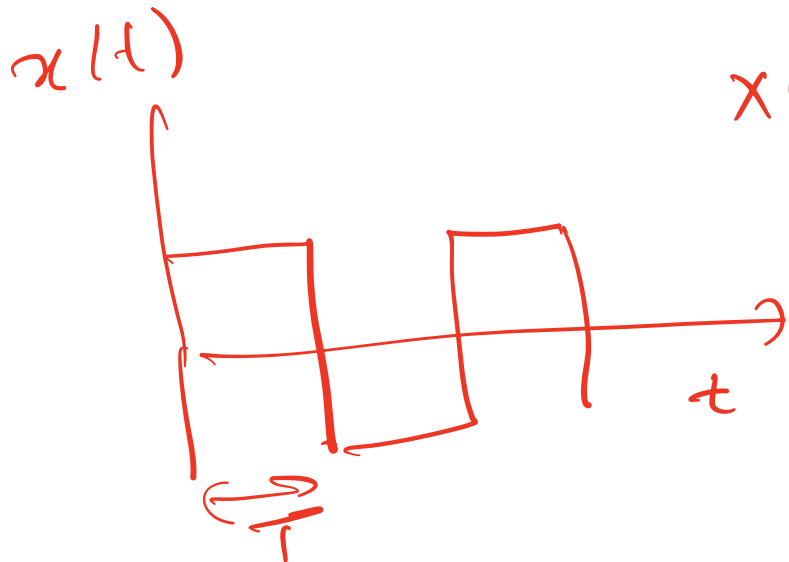
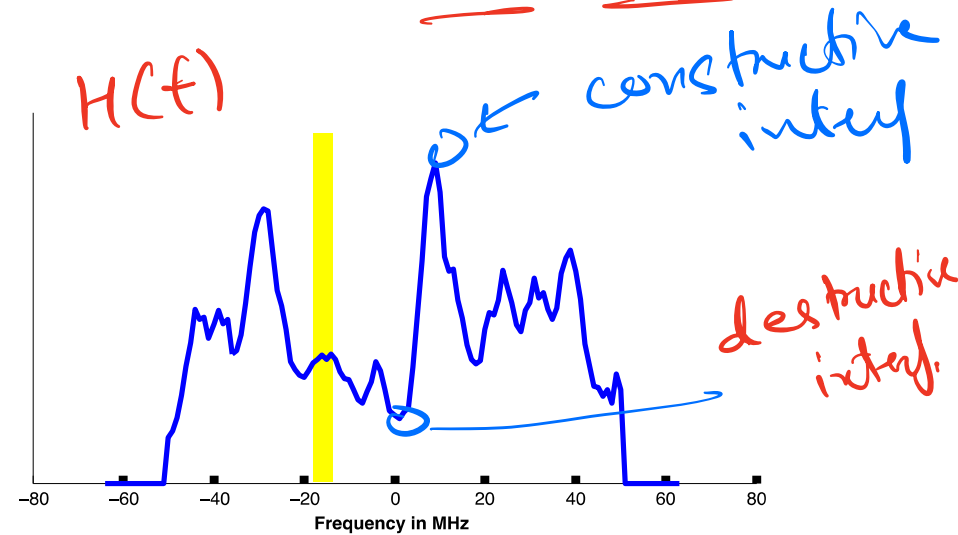
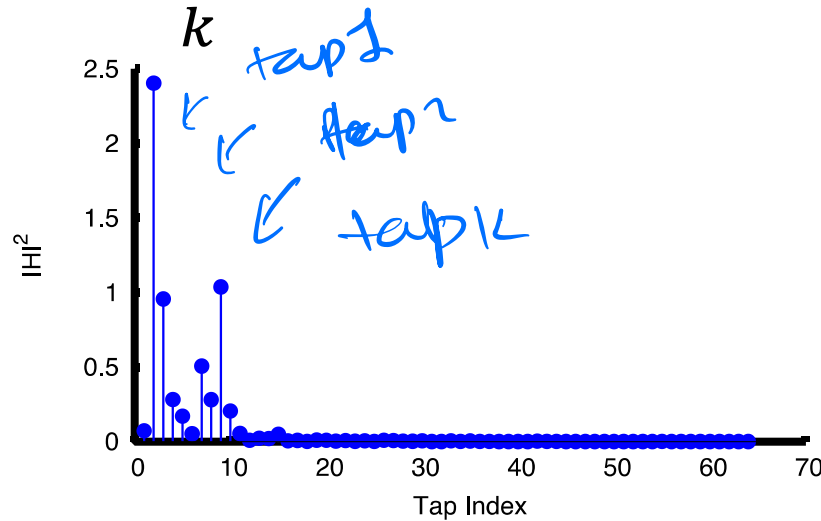


$$h(t) = \sum_{k=1}^L h_k \delta(t - \tau_k)$$

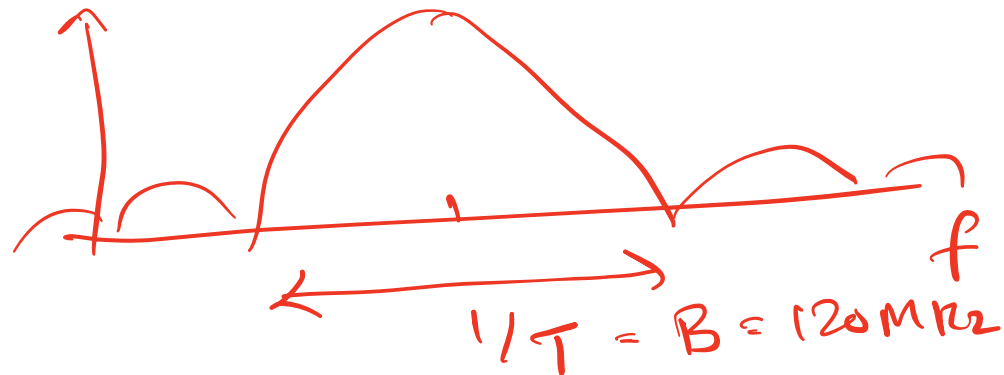


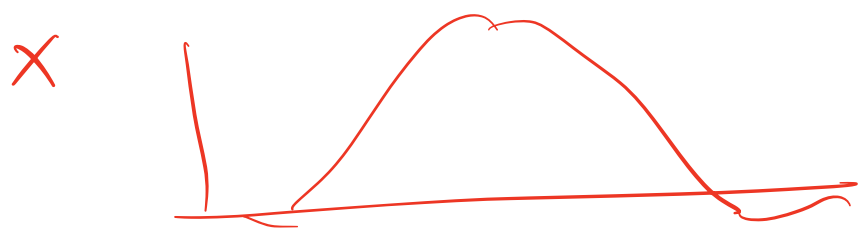
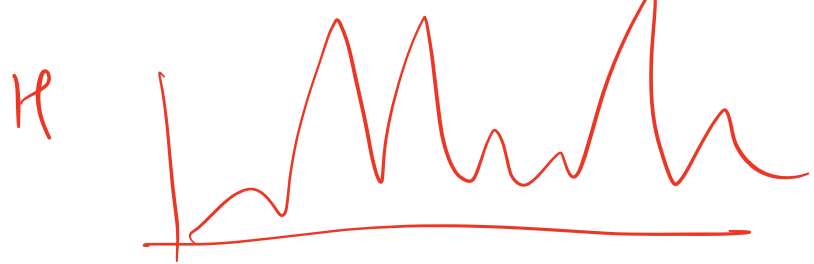
Wide Band vs Narrow Band Channel

$$y(t) = \sum_k h_k x(t - \tau_k) = h(t) * x(t) \Leftrightarrow \underline{H(f)} \underline{X(f)}$$

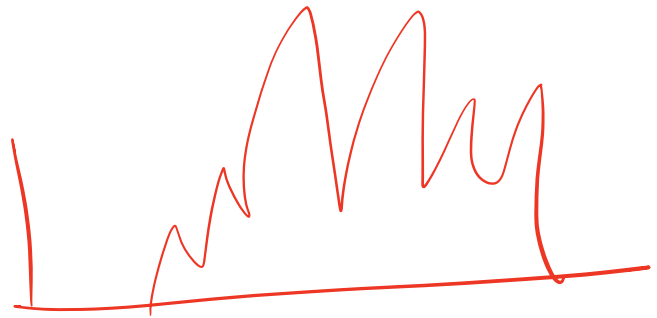


$X(f)$





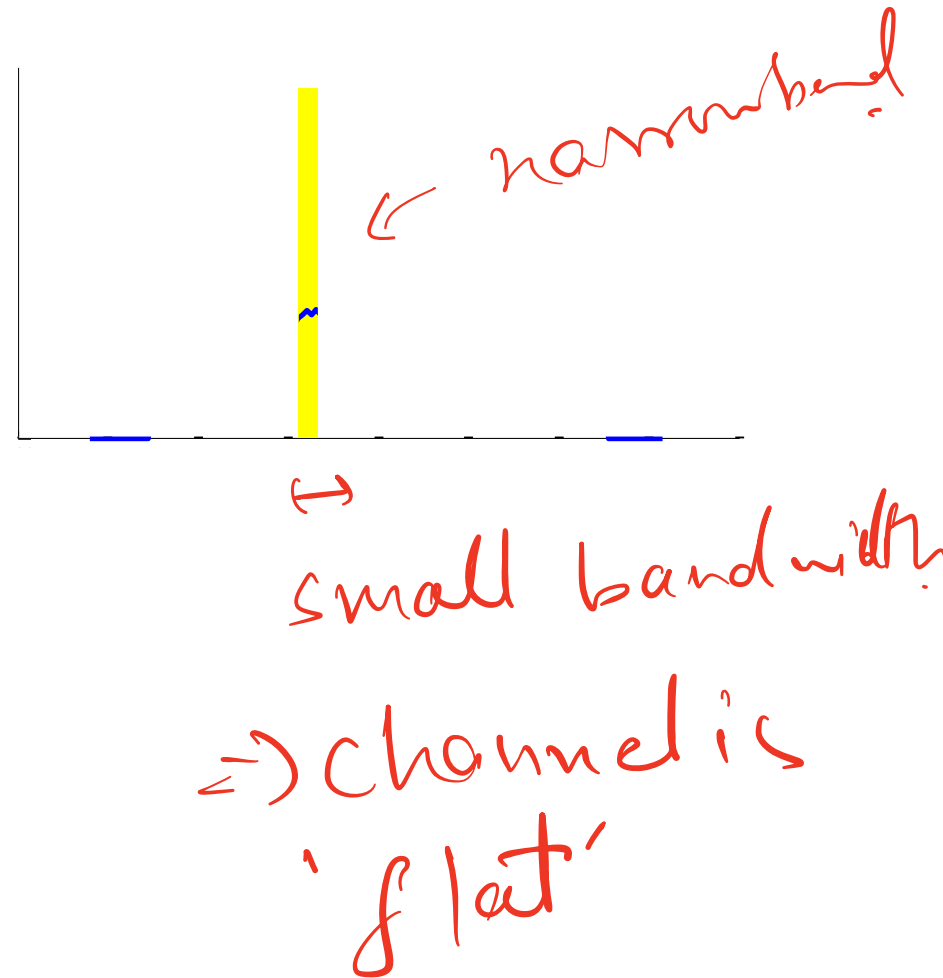
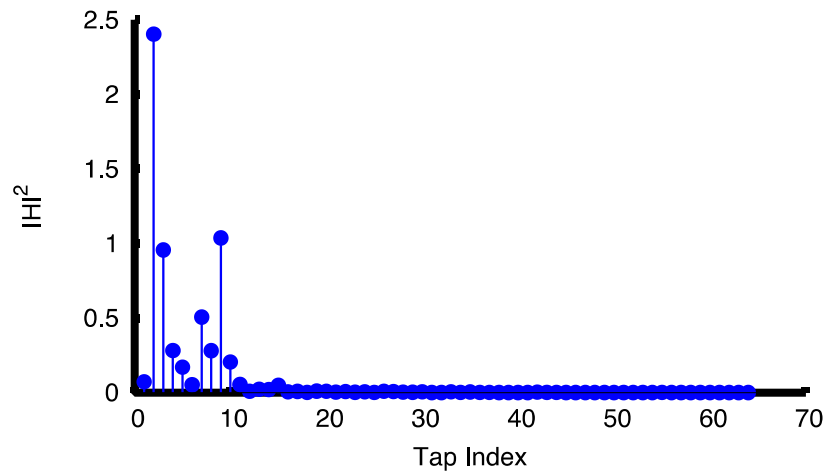
$$Y \equiv XH$$



distortion $\downarrow$
due to channel $\Rightarrow$
cannot recover
tx data 'X'

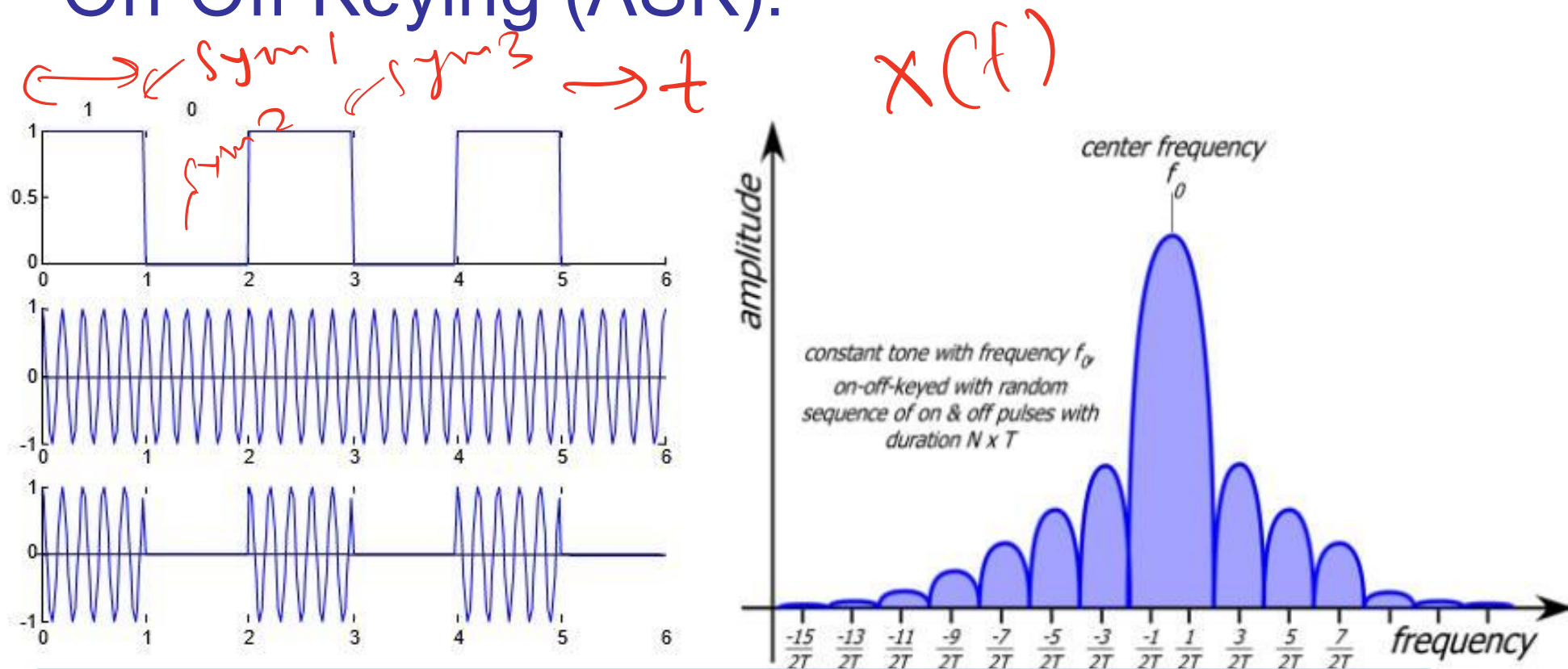
Wide Band vs Narrow Band Channel

$$y(t) = \sum_k h_k x(t - \tau_k) = h(t) * x(t) \Leftrightarrow H(f)X(f)$$



Digital Modulation

- Map Bits to Signal Values
- On-Off Keying (ASK):



Suffers from Frequency-selective channel $\rightarrow$ ISI

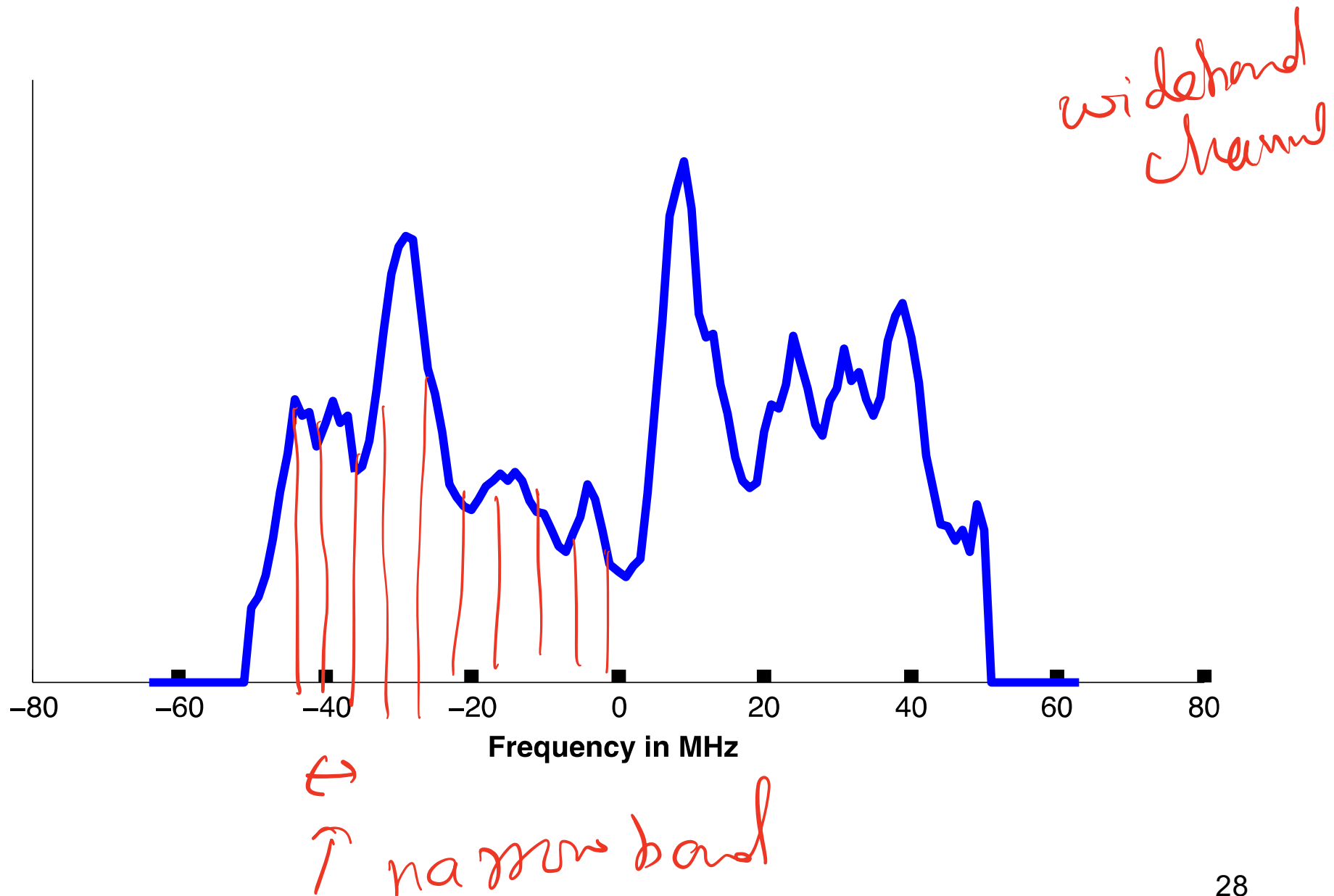
- Solution:

OFDM: Orthogonal Frequency Division Multiplexing



- Idea: transmit symbols in frequency not time.

Transmit In Frequency Domain



Orthogonality

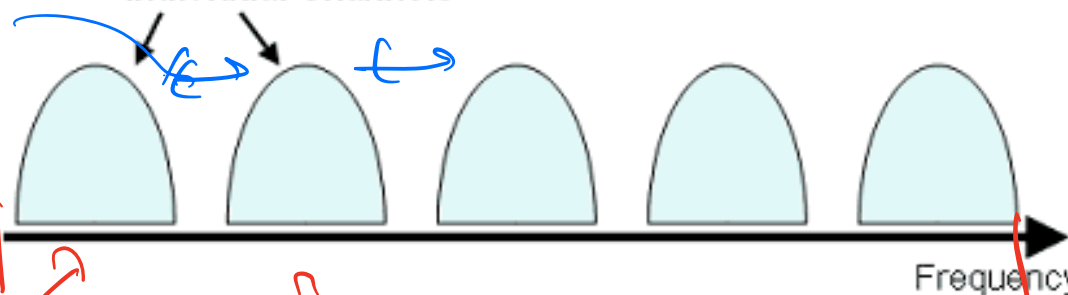
guard band

freq Division multiplexing

FDM

$$B = \frac{1}{T}$$

Individual Channels



sym 1

sym 2

OFDM

B

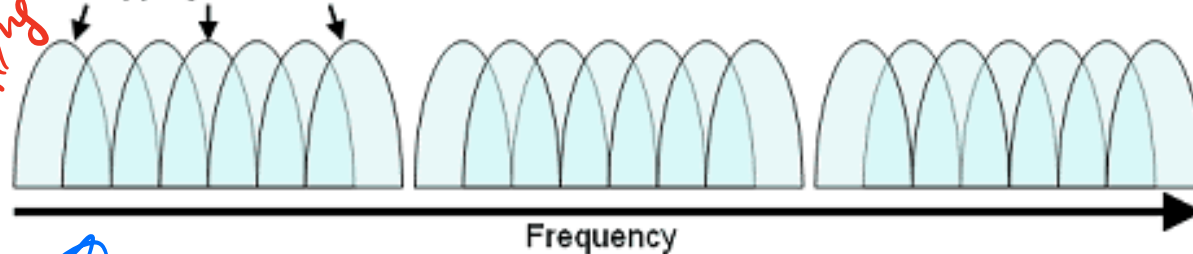
freq carrier

Channel 0

Channel 1

Channel 2

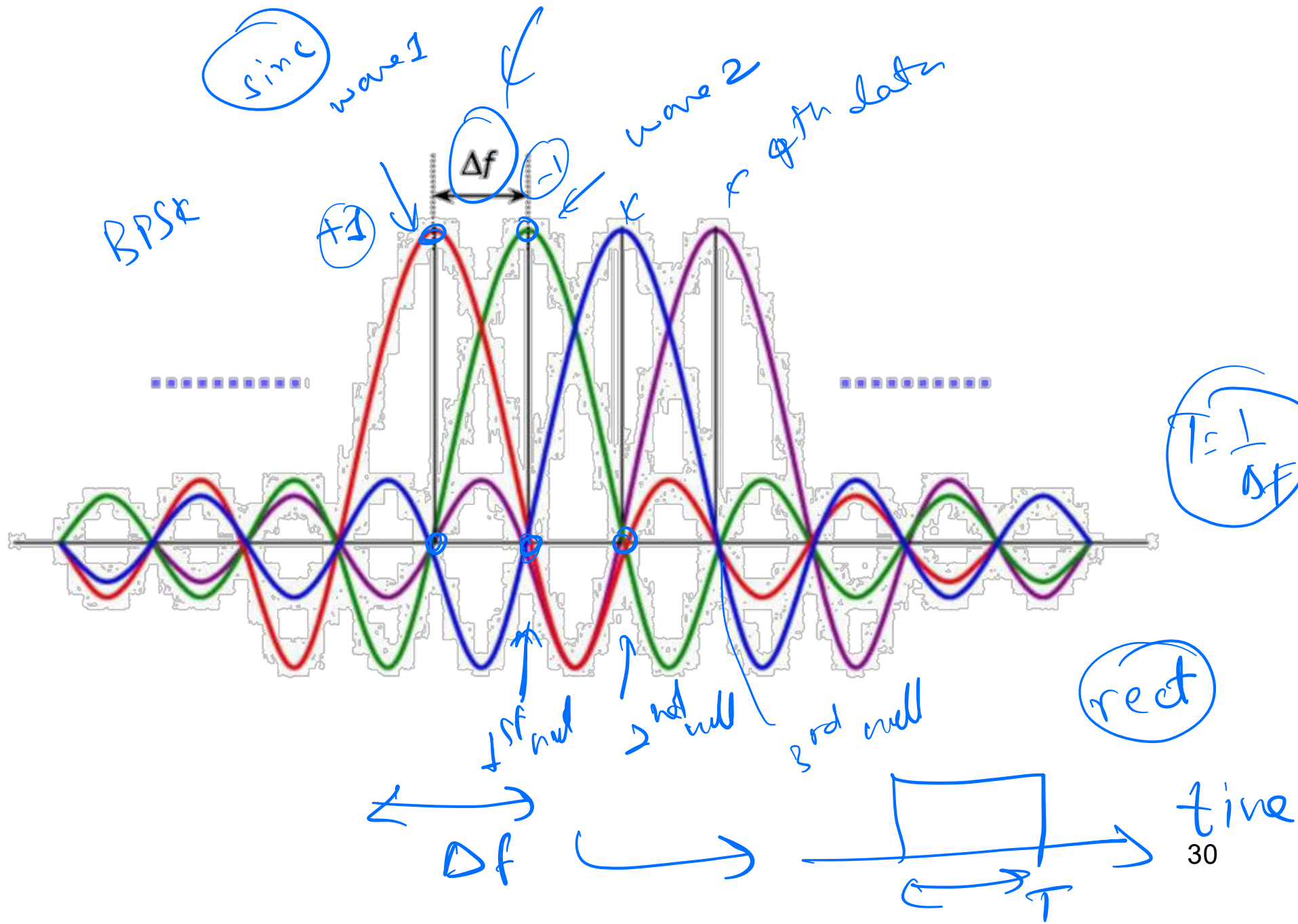
Overlapping Sub-Carriers

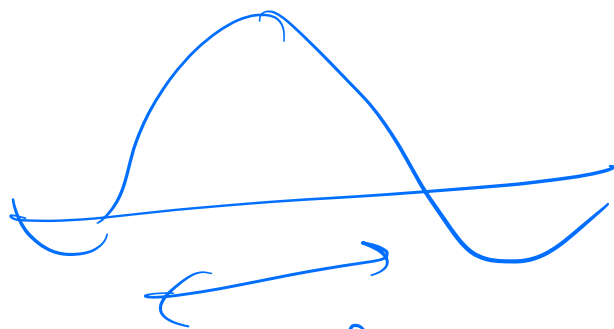


Orthogonal
freq
division
multiplexing

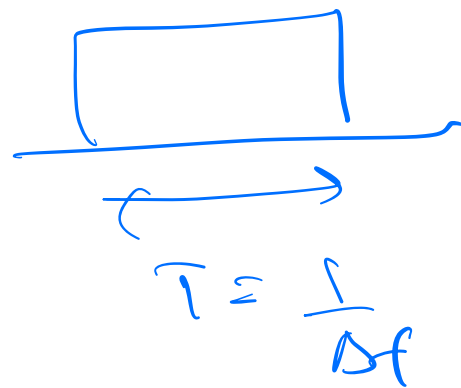
Orthogonality

diff between two bins

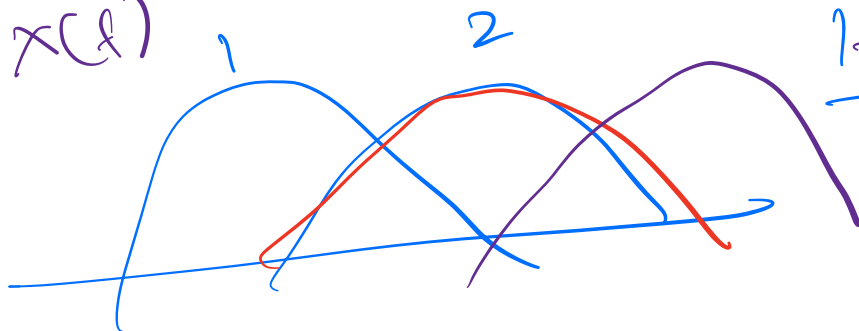




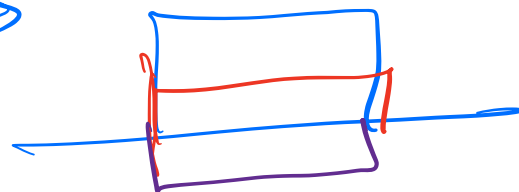
$\xrightarrow{1/f}$



$x(f)$

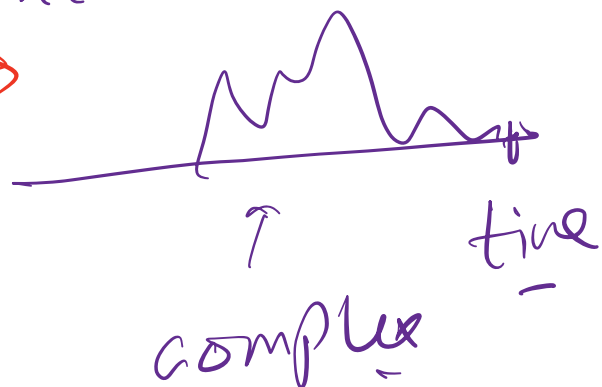


$\xrightarrow{HFT}$

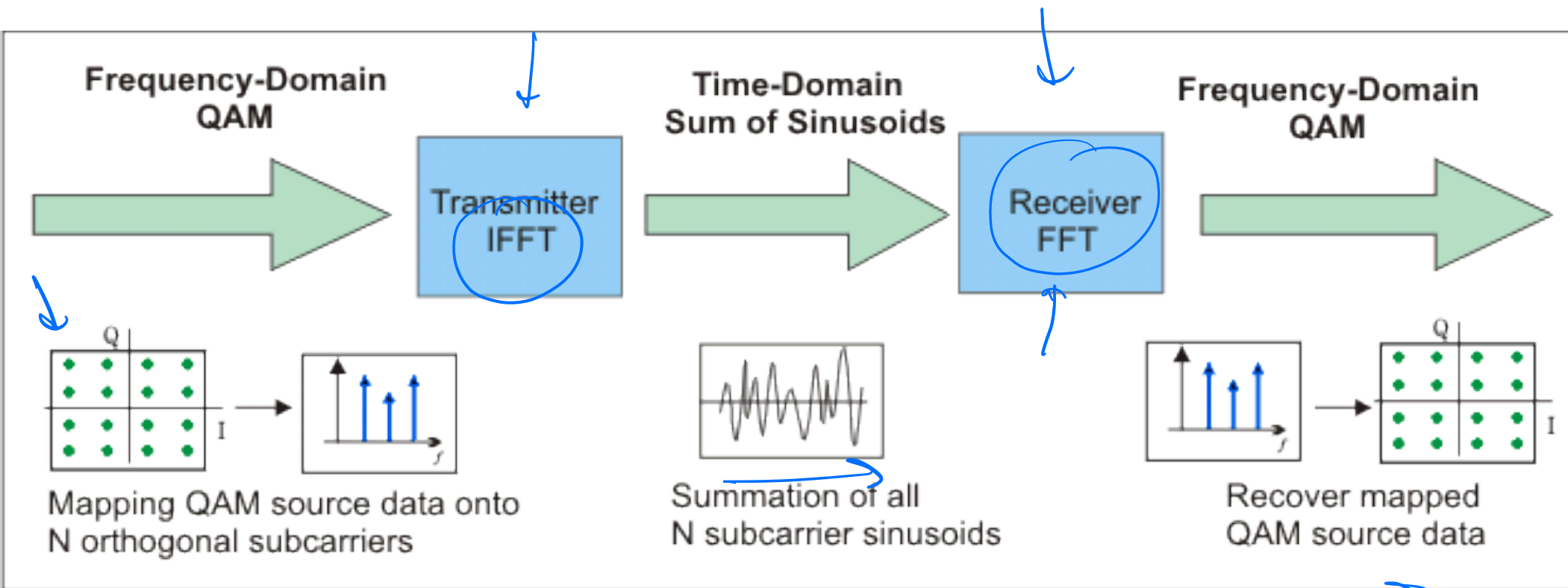


$x(t)$

$\rightarrow$



Orthogonal Frequency Division Multiplexing



Simplified OFDM System Block Diagram

bit to symbol

symbol to bit decoding

Discrete Fourier Transform

$$x(t) \Rightarrow X(f) = \frac{1}{N} \sum_{k=1}^N x(k) e^{-j 2 \pi f \frac{k}{N}}$$

IDFT

$$X(0) = \frac{1}{N} \sum x(t)$$

$$\text{Bits} \Rightarrow \text{modulate } x(t) \Rightarrow X(f) \Rightarrow x(t) = \sum_{f=1}^N X(f) e^{j 2 \pi f \frac{t}{N}}$$

• Band width $B \Rightarrow \Delta f = \frac{B}{N}$ bandwidth

N symbols over
Bandwidth
 B

width of bin

$$\Delta f = \frac{1}{T}$$

$\Rightarrow B$ sample/sec

$$\text{Time} = \frac{N}{B} = T$$

freq down

Why OFDM always requires complex IQ?

